Girls Build Bots — Cadette Cybersecurity Investigator

Badge-aligned workshop guide for robotics-team-led outreach

Level: Cadettes (Grades 6–8) · **Badge:** Cybersecurity — Cybersecurity Investigator **Series:** Cybersecurity (Basics · Safeguards · Investigator) **Format:** One 2-hour session · **Ratio:** 1 mentor per 5–6 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — everything runs on a mentor-prepared (entirely fictional) case file.

> **Troop Leaders:** This curriculum is designed to *align with* the Cadette Cybersecurity badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

The through-line: understanding, protecting, investigating

The capstone: Cadettes run a full incident investigation — thinking like both the attacker (how was this done?) and the defender (how do we prove it and prevent it?). The discipline is evidence-first reasoning: no conclusion outruns its support.

Vocabulary: incident, evidence chain, timeline, attack vector, hypothesis, corroborate, attribution, remediation.

Cybersecurity Investigator (2 hours)

Badge theme: Investigating a (fictional) cyber incident end to end — tracing the attack, building an evidence chain, and recommending defenses.

Materials

- **The case file** (mentor-prepared, wholly fictional — invented names, no real brands): "The Bake Sale Account Takeover." A club's social account posted spam overnight. Evidence packets: (1) a phishing DM sent to the club treasurer three days prior, (2) a login-history sheet showing an unfamiliar location at 2:14 AM, (3) the treasurer's statement ("I clicked a link to 'verify' our account"), (4) a screenshot of the fake login page (subtly wrong URL), (5) a red-herring packet — a rival club member who "seemed suspicious" but whose evidence doesn't hold up
- Evidence log sheets (Exhibit / What it shows / Corroborated by / Confidence: high-med-low)
- Timeline strips + a wall or table to build on; incident-report template for the final brief

Activity 1 — How Real Investigations Work (15 min)

The method professionals use: secure the evidence, build a **timeline**, form **hypotheses**, test each against the evidence, and only then conclude. Two rules Cadettes adopt for the day: every claim needs an exhibit, and the goal is the *attack vector* (how it happened), not just a culprit. Warning delivered with a smile: the file contains a red herring, because real cases always do.

Activity 2 — Work the Case (50 min)

Teams of 3–4 receive the full file and work it: 1. **Log every exhibit** with what it shows and its confidence level 2. **Build the timeline** physically with the strips (DM → click → fake login → 2:14 AM access → spam posts) 3. **Form hypotheses** — at least two, including the red-herring suspect — and test each: which exhibits support it? which contradict it? 4. **Identify the attack vector** (phishing link → credential theft → account takeover) with the evidence chain that proves it

Mentors float as "senior investigators": answer factual questions from a prepared key, never confirm conclusions. The red herring's collapse under evidence is the session's teachable moment — suspicion isn't proof.

Activity 3 — The Incident Report (30 min)

Each team completes the report template like professionals: what happened (timeline), how (attack vector + evidence chain), what we can't prove (unknowns stated honestly — real reports always have them), and **remediation** — three defenses that would have stopped this (2FA, URL-checking habit, verify-before-clicking). The remediation section ties the whole badge series together: Basics explained the system, Safeguards taught the defenses, Investigator shows exactly where they'd have worked.

Activity 4 — Case Presentations (20 min) + Wrap (5 min)

Teams brief their findings in five minutes each; the audience gets one cross-examination question per team ("how do you know the login wasn't the treasurer traveling?" — teams must answer from exhibits). Closing: each Scout names one investigative habit she's keeping — evidence before conclusions, corroborate everything, or beware the obvious suspect.

> **Robot moment (optional):** Share for 60 seconds about a time the team debugged a mystery robot failure the same way — logs, timeline, hypotheses tested one at a time — and how the first suspect is often innocent.

Completion Checklist — Cybersecurity Investigator

By the end of the session, to earn the badge, each Scout can: - [ ] Build an evidence log and timeline from an incident file - [ ] Test multiple hypotheses against evidence and reject one that fails - [ ] Identify an attack vector and support it with an evidence chain - [ ] Write remediation recommendations and defend conclusions under questioning

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Mentor Guide (for the robotics team)

- **The case file is everything.** Invest 45 minutes of prep; keep it wholly fictional (invented club, fake usernames, no real brands or people). A well-made red herring is the best teaching tool in the session.
- **You don't need a robot for this.** The "robot moment" is a 60-second add-on.
- **Answer facts, never verdicts.** "What does exhibit 2 show?" gets an answer; "so it was phishing, right?" gets "what's your evidence?"
- **Honor the unknowns.** Praise teams that say "we can't prove X" — epistemic honesty is the mark of a real investigator.
- **Keep cross-examination respectful and evidence-bound.** One question per team, answered from exhibits — rigor without pile-ons.

📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Cadette Cybersecurity badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.